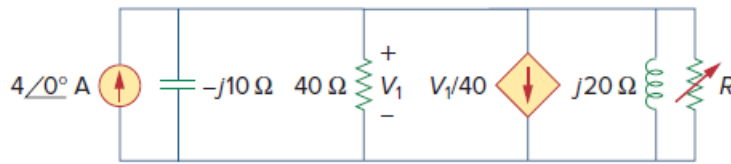


Problem

The load resistance R_L in Fig. 11.51 is adjusted until it absorbs the maximum average power. Calculate the value of R_L and the maximum average power.

Figure 11.51:



Step-by-step solution

Step 1 of 9

Refer to Figure 11.51 in the text book.

Consider the circuit diagram shown in Figure 1 to calculate the open circuit voltage.

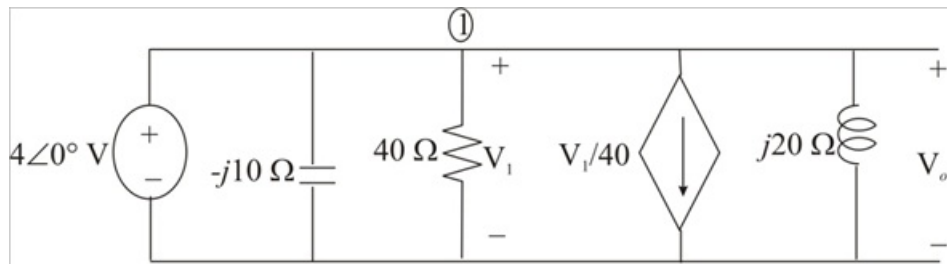


Figure 1

Step 2 of 9

Apply Kirchhoff's current law at node 1.

$$\frac{V_1}{40} + \frac{V_1}{-j10} + \frac{V_1}{j20} + \frac{V_1}{40} - 4\angle 0^\circ = 0$$

$$V_1 \left(\frac{1}{40} + \frac{j}{10} - \frac{j}{20} + \frac{1}{40} \right) = 4\angle 0^\circ$$

$$V_1 (0.025 + j0.1 - j0.05 + 0.025) = 4\angle 0^\circ$$

$$V_1 (0.05 + j0.05) = 4\angle 0^\circ$$

$$V_1 = \frac{4\angle 0^\circ}{0.0707\angle 45^\circ} \\ = 56.577\angle -45^\circ \text{ V}$$

Therefore, the value the voltage V_1 is, $56.577\angle -45^\circ \text{ V}$.

The value of the open circuit voltage is equal to voltage V_1 that is, $56.577\angle -45^\circ \text{ V}$.

Step 3 of 9

Short the output terminals as shown in Figure 2 to calculate the short circuit current.

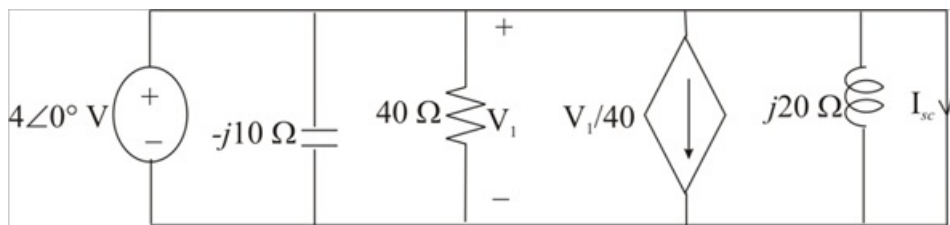


Figure 2

Step 4 of 9

Notice that all the elements are shorted. Hence, the current from the source completely flows through the shorted wire. Therefore, the value of the short circuited current I_{sc} is, $4\angle 0^\circ \text{ A}$.

Step 5 of 9

Calculate the value of the Thevenin's impedance Z_{Th} .

$$Z_{Th} = \frac{V_{oc}}{I_{sc}}$$

Substitute $4\angle 0^\circ \text{ A}$ for I_{sc} and $56.577\angle -45^\circ \text{ V}$ for V_{oc} .

$$\begin{aligned} Z_{Th} &= \frac{56.577\angle -45^\circ}{4\angle 0^\circ} \\ &= 14.144\angle -45^\circ \\ &= 10 - j10 \Omega \end{aligned}$$

Therefore, the value of the Thevenin's impedance Z_{Th} is, $10 - j10 \Omega$.

Step 6 of 9

Calculate the value of the load resistance R_L that will absorb the maximum power.

$$R_L = |Z_{Th}|$$

Substitute $10 - j10 \Omega$ for Z_{Th} .

$$\begin{aligned} R_L &= |10 - j10| \\ &= \sqrt{(10)^2 + (10)^2} \\ &= \sqrt{100 + 100} \\ &= \sqrt{200} \end{aligned}$$

$$R_L = 14.14 \Omega$$

Therefore, the value of the load resistance R_L that will absorb the maximum power is,

$$\boxed{14.14 \Omega}.$$

Step 7 of 9

Draw the Thevenin's equivalent circuit diagram.

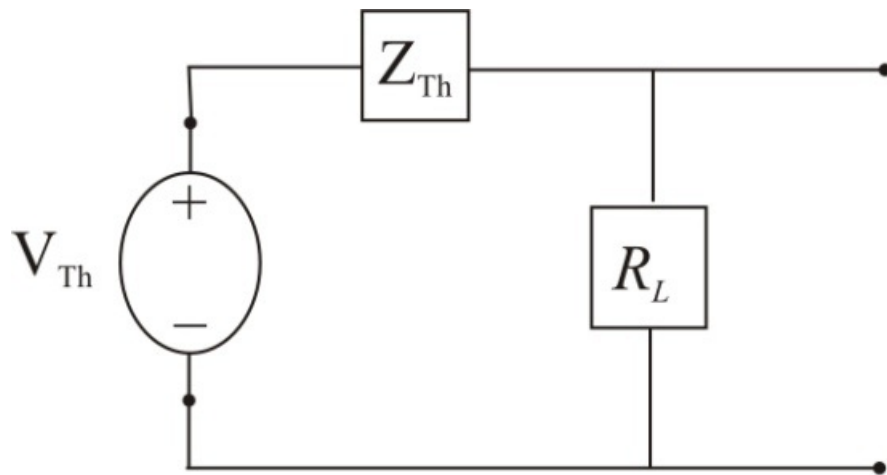


Figure 3

Step 8 of 9

Calculate the value of the current I flowing in the circuit.

$$I = \frac{V_{Th}}{Z_{Th} + R_L}$$

Substitute $56.577 \angle -45^\circ$ V for V_{Th} , $10 - j10 \Omega$ for Z_{Th} and 14.14Ω for R_L .

$$\begin{aligned} I &= \frac{56.577 \angle -45^\circ}{10 - j10 + 14.14} \\ &= \frac{56.577 \angle -45^\circ}{24.14 - j10} \\ &= \frac{56.577 \angle -45^\circ}{26.157 \angle -22.48^\circ} \\ &= 2.163 \angle -22.52^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I flowing in the circuit is, $2.163 \angle -22.52^\circ$ A.

Step 9 of 9

Calculate the value of the maximum average power P_{max} absorbed by the load resistance.

$$P_{max} = \frac{1}{2} |I|^2 R_L$$

Substitute 2.163 A for $|I|$ and 14.14Ω for R_L .

$$\begin{aligned} P_{max} &= \frac{1}{2} (2.163)^2 (14.14) \\ &= (4.679)(7.07) \\ &= 33.08 \text{ W} \end{aligned}$$

Therefore, the value of the maximum average power P_{max} absorbed by the load resistance is,

$$\boxed{33.08 \text{ W}}$$